MATH317 · NUMERICAL METHODS

# Midterm 472 تدريب

12/04/2026 · 60 دقيقة · 30 درجة

**محارب**

## 01 الأسئلة

حل كل سؤال في المساحة اللي تحته. الحل الكامل في ملف منفصل.

Q1 TRUE / FALSE · 7 M

**1) Significant digits are only important when rounding final results, not during intermediate calculations.**
**2) Horner's algorithm is used for efficiently evaluating polynomials.**
**3) The Taylor series converges rapidly for points far from the expansion point.**
**4) The Bisection method guarantees finding the exact root in a finite number of steps.**
**5) Rounding replaces a number with an approximation that minimizes error, whereas chopping discards digits without adjustment.**
**6) The Mean-Value Theorem is a special case of Taylor's Theorem with zero derivatives.**
**7) The error in the Bisection method after n steps is at most (b – a) divided by $2^n$.**

3 + 5 M **Q2**

Given $f(x) = 22 - 4x^2 + 3x^3$.

a) Use Horner's algorithm to evaluate f(x) at x = −2.

b) What is the Taylor series for the function f(x) about c = −3?

7 M **Q3**

Determine the single-precision machine representation in a 32-bit word-length computer of the decimal number (−23.21875).

2 + 6 M

**Q4**

Given $f(x) = -x^2 + 4x - e^x$.

a) Prove that f(x) has exactly one real root α in the interval [0, 1].

b) Find a root of f(x) on [0, 1] by using the bisection method. Terminate the iterations if they exceed nmax = 2 or if the error bound is less than $0.5 \times 10^{-6}$.